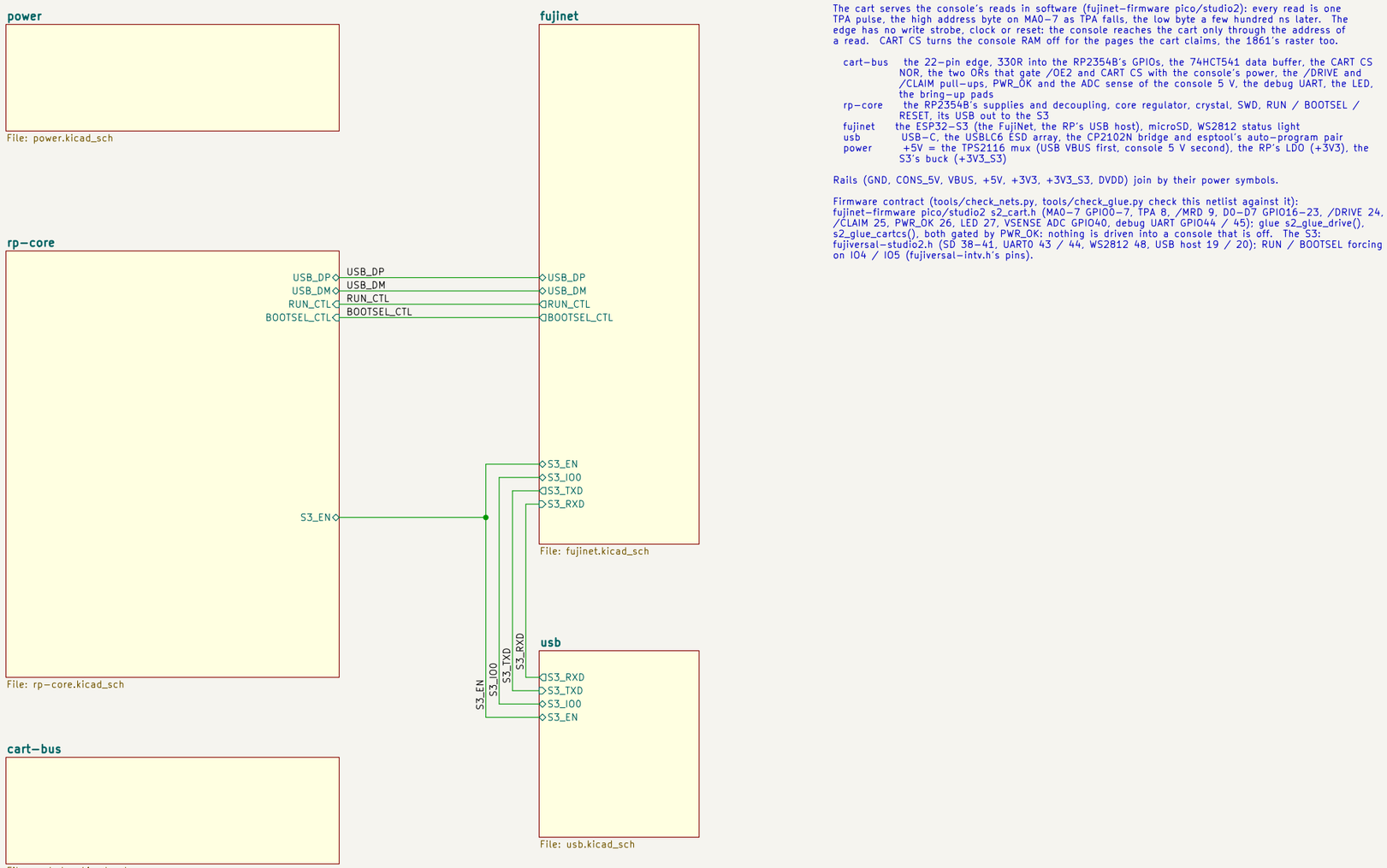


the console cartridge slot (the edge) is on cart-bus

FujiNet for the RCA Studio II, Rev0: an RP2354B cartridge on the CDP1802 bus + ESP32-S3.

The cart serves the console's reads in software (fujinet-firmware pico/studio2): every read is one TPA pulse, the high address byte on MA0-7 as TPA falls, the low byte a few hundred ns later. The edge has no write strobe, clock or reset: the console reaches the cart only through the address of a read. CART CS turns the console RAM off for the pages the cart claims, the 1861's raster too.

cart-bus	the 22-pin edge, 330R into the RP2354B's GPIOs, the 74HCT541 data buffer, the CART CS NOR, the two ORs that gate /OE2 and CART CS with the console's power, the /DRIVE and /CLAIM pull-ups, PWR_OK and the ADC sense of the console 5 V, the debug UART, the LED, the bring-up pads
rp-core	the RP2354B's supplies and decoupling, core regulator, crystal, SWD, RUN / BOOTSEL / RESET, its USB out to the S3
fujinet	the ESP32-S3 (the FujiNet, the RP's USB host), microSD, WS2812 status light
usb	USB-C, the USBLC6 ESD array, the CP2102N bridge and esptool's auto-program pair
power	+5V = the TPS2116 mux (USB VBUS first, console 5 V second), the RP's LDO (+3V3), the S3's buck (+3V3_S3)

Rails (GND, CONS_5V, VBUS, +5V, +3V3, +3V3_S3, DVDD) join by their power symbols.

Firmware contract (tools/check_nets.py, tools/check_glue.py check this netlist against it): fujinet-firmware pico/studio2 s2_cart.h (MA0-7 GPIO0-7, TPA 8, /MRD 9, D0-D7 GPIO16-23, /DRIVE 24, /CLAIM 25, PWR_OK 26, LED 27, VSENSE ADC GPIO40, debug UART GPIO44 / 45); glue s2_glue_drive(), s2_glue_cartcs(), both gated by PWR_OK: nothing is driven into a console that is off. The S3: fujiversal-studio2.h (SD 38-41, UART0 43 / 44, WS2812 48, USB host 19 / 20); RUN / BOOTSEL forcing on IO4 / IO5 (fujiversal-intv.h's pins).

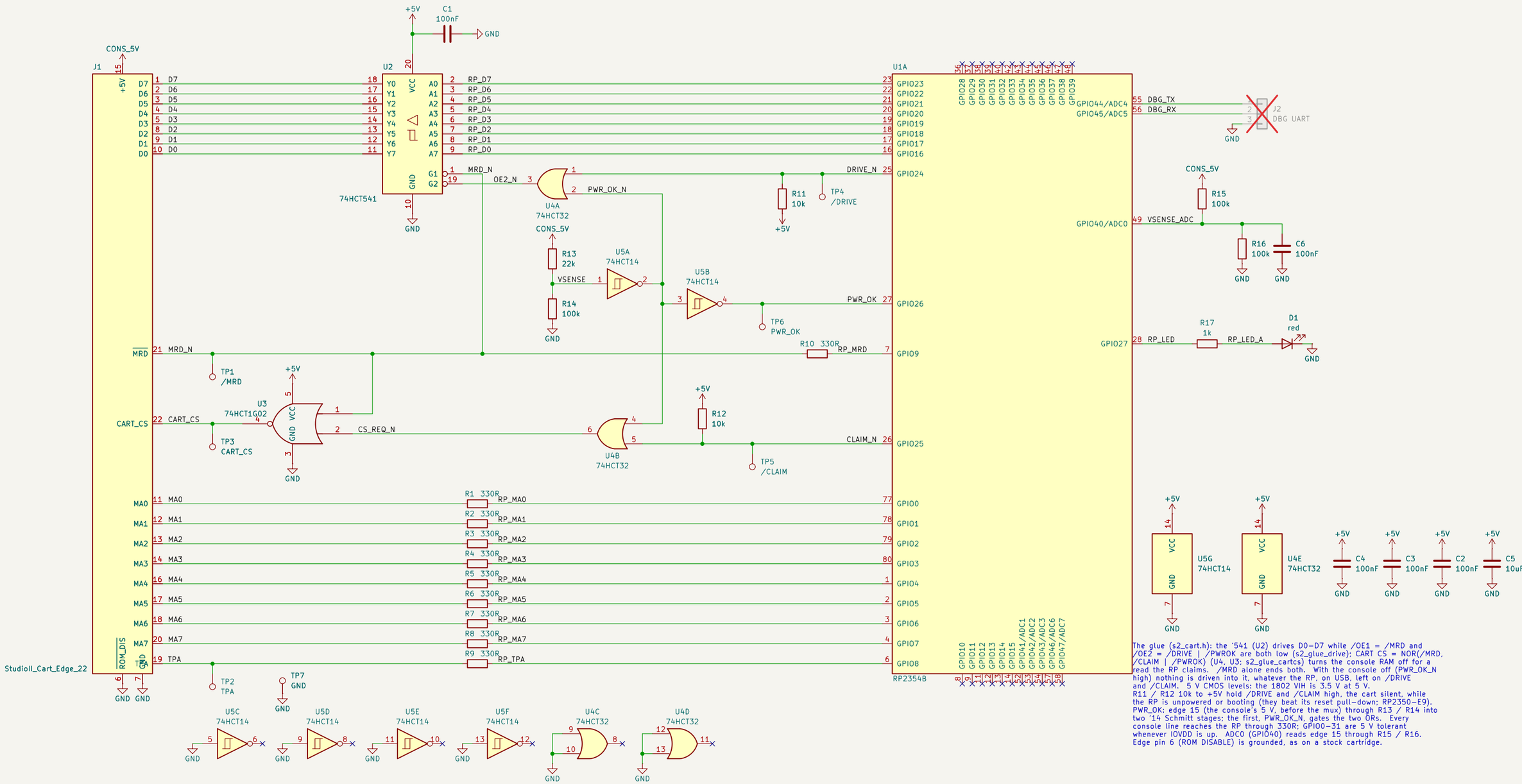
Rev0 2026-10-11: RP2354B + ESP32-S3; firmware pico/studio2 s2_cart.h, fujiversal-studio2.h
CERN-OHL-W-2.0 (RP core from FujiNet-5200-Rev0, S3 half from FujiNet-ChannelF-Rev0)
Generated from tools/design.py + tools/sch_layout.py - edit those, not this file

FujiNet

Sheet: /
File: FujiNet-StudioII-Rev0.kicad_sch

Title: FujiNet Studio II Rev0

Size: A3	Date: 2026-10-11	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 1/6



Rev0 2026-10-11: RP2354B + ESP32-S3; firmware pico/studio2 s2_cart.h, fujiversal-studio2.h
CERN-OHL-W-2.0 (RP core from FujiNet-5200-Rev0, S3 half from FujiNet-ChannelF-Rev0)
Generated from tools/design.py + tools/sch_layout.py — edit those, not this file

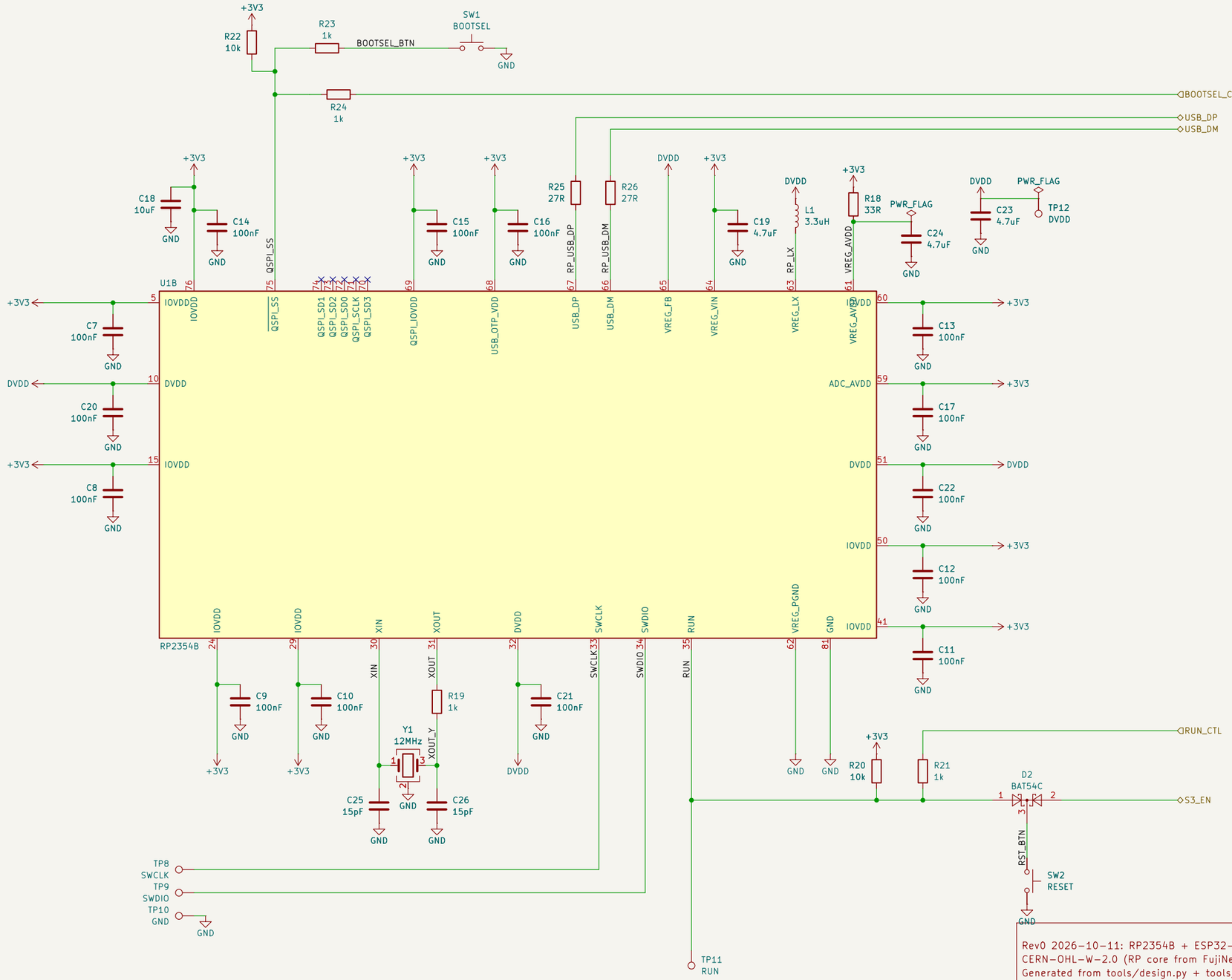
FujiNet

Sheet: /cart-bus/
File: cart-bus.kicad_sch

Title: FujiNet Studio II Rev0 — Cart bus: edge, glue, RP GPIO

Size: A3	Date: 2026-10-11	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 2/6

Supplies (RP2350 datasheet 6.3.7): IOVDD, QSPI_IOVDD, USB_OTP_VDD, ADC_AVDD and VREG_VIN on +3V3, the AP2112K LDO that rises with +5V; the core regulator switches VREG_LX through 3.3 uH onto DVDD (VREG_FB senses it), its analog supply VREG_AVDD through 33R / 4.7 uF. One 100 nF at every supply pin, drawn at the pin it serves. RESET resets the RP2354B and the ESP32-S3 together (D2 steers RST_BTN onto RUN and S3_EN): the way back to the FujiNet's CONFIG. Hold BOOTSEL while pressing RESET for the RP's USB boot ROM; the S3 does both itself: IO4 (RUN_CTL) low resets the RP, IO5 (BOOTSEL_CTL) low through that reset selects BOOTSEL. USB: 27R at the RP, the S3 is the host. SWD and RUN on bring-up pads. QSPI_SD0-3 / SCLK unused: the 2 MB flash is in the package.



Rev0 2026-10-11: RP2354B + ESP32-S3; firmware pico/studio2 s2_cart.h, fujiversal-studio2.h
CERN-OHL-W-2.0 (RP core from FujiNet-5200-Rev0, S3 half from FujiNet-ChannelF-Rev0)
Generated from tools/design.py + tools/sch_layout.py - edit those, not this file

FujiNet

Sheet: /rp-core/
File: rp-core.kicad_sch

Title: FujiNet Studio II Rev0 - RP2354B core, crystal, RUN, SWD

Size: A3	Date: 2026-10-11	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 3/6

ESP32-S3-WROOM-1-N16R8: the FujiNet (fujiversal-studio2.h pin map), on +3V3_S3 (the buck).
EN: 10k / 1 uF power-on delay. The S3_RST button, the cart RESET
(rp-core, through the BAT54C) and the USB bridge's auto-program pair
(usb sheet) all pull it low; S3_BOOT and the same pair pull IO0.
IO4 / IO5 drive the RP2354B's RUN and BOOTSEL (fujiversal-intv.h's
pins); IO19 / IO20, the S3's own USB, host the RP's.
microSD in SPI mode, 10k pull-ups on CS, DO, DAT1, DAT2 and CD.
WS2812B-2020 on +5V: its data input (VIH 2.7 V) takes the S3's 3.3 V.
The 23 spare GPIOs (right) are not connected.

+3V3_S3
R27
10k
S3_EN
SW3
S3_RST
GND
C29
1uF
GND
+3V3_S3
C28
100nF
GND
C27
22uF
GND
U6
EN
3V3
S3_TXD
S3_RXD
TXD0
RXD0
S3_IO0
IO0
SW4
S3_BOOT
GND
RUN_CTL
BOOTSEL_CTL
IO4
IO5
USB_DP
USB_DM
USB_D+
USB_D-
IO1
IO2
IO3
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO21
IO35
IO36
IO37
IO45
IO46
IO47
IO48
GND
IO42
IO41
IO40
IO39
IO38
ESP32-S3-WROOM-1-N16R8
LED_STRIP
GND
+3V3_S3
RN1C
4x10k
+3V3_S3
+3V3_S3
+3V3_S3
+3V3_S3
R28
10k
J3
SD_DAT1
SD_DAT2
SD_MOSI
SD_SCK
SD_MISO
SD_CS
SD_CD
DAT1
DAT2
CMD/DI
CLK
DAT0/DO
DAT3/CS
CD
VDD
VSS
microSD
+3V3_S3
GND
C30
10uF
GND
+5V
C31
100nF
GND
DOUT
VDD
DIN
VSS
WS_DIN
R29
330R
D3
WS2812B-2020-V6
GND

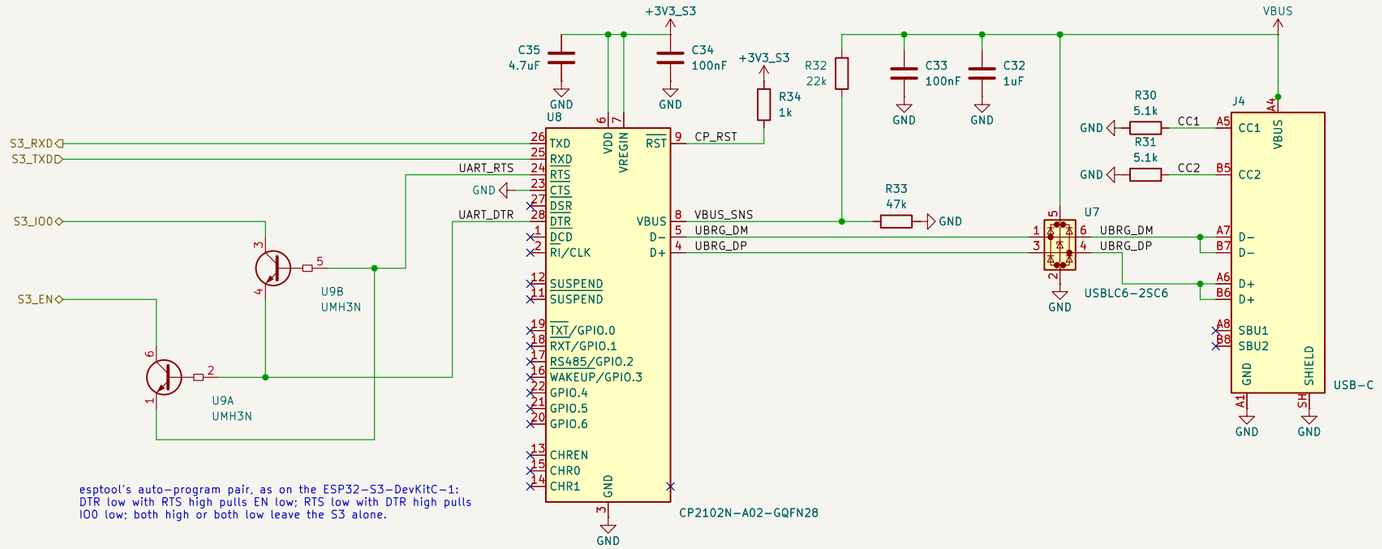
Rev0 2026-10-11: RP2354B + ESP32-S3; firmware pico/studio2 s2_cart.h, fujiversal-studio2.h
CERN-OHL-W-2.0 (RP core from FujiNet-5200-Rev0, S3 half from FujiNet-ChannelF-Rev0)
Generated from tools/design.py + tools/sch_layout.py — edit those, not this file

FujiNet

Sheet: /fujinet/
File: fujinet.kicad_sch

Title: FujiNet Studio II Rev0 — FujiNet (ESP32-S3), microSD

Size: A3	Date: 2026-10-11	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 4/6



esptool's auto-program pair, as on the ESP32-S3-DevKitC-1:
DTR low with RTS high pulls EN low; RTS low with DTR high pulls
IO0 low; both high or both low leave the S3 alone.

Rev0 2026-10-11: RP2354B + ESP32-S3; firmware pico/studio2 s2_cart.h, fujiversal-studio2.h
CERN-OHL-W-2.0 (RP core from FujiNet-5200-Rev0, S3 half from FujiNet-ChannelF-Rev0)
Generated from tools/design.py + tools/sch_layout.py — edit those, not this file

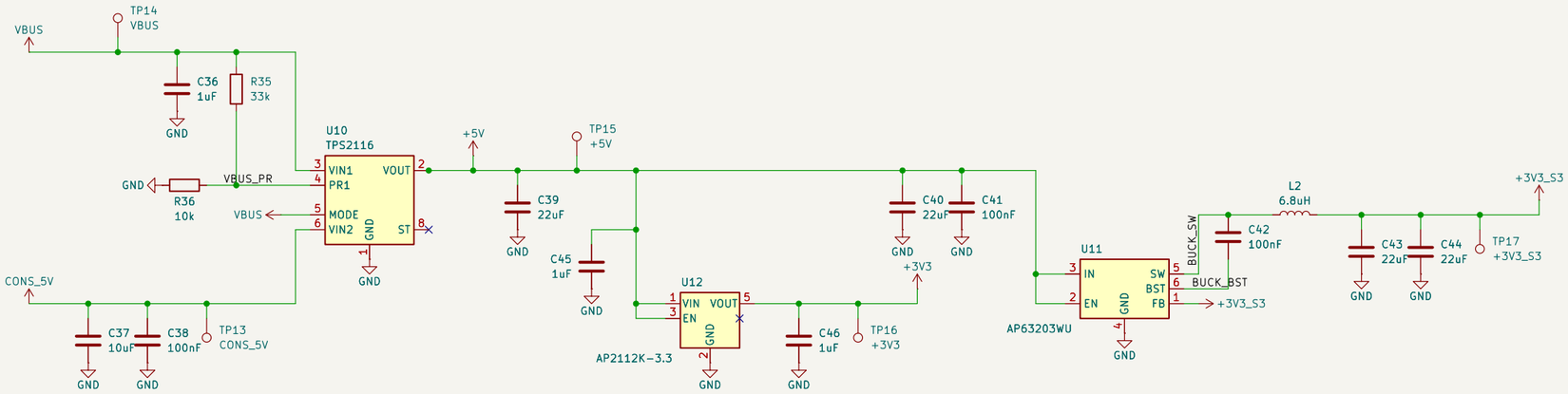
FujiNet

Sheet: /usb/
File: usb.kicad_sch

Title: FujiNet Studio II Rev0 — USB-C + CP2102N

Size: A3	Date: 2026-10-11	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 5/6

Power tree: +5V is the TPS2116 power mux's output (U10), USB first, the console's 5 V second.
  VBUS   USB-C (usb sheet) -> VIN1, the priority input: MODE on VIN1, PR1 from R35 33k / R36 10k, so VBUS
            above VREF x 4.3 (3.96-4.64 V) runs the cart and the console's 7805 supplies nothing.
  CONS_5V edge 15, the console's 7805 -> VIN2, whenever VBUS is absent.
  37-60 mOhm either way: +5V sits within ~20 mV of its source, so the 74HCT glue stays inside 4.5-5.5 V
  on either; reverse current blocking: neither source back-feeds the other. ST is not used.
+5V feeds the 74HCT541 / '32 / '1G02 / '14, the WS2812, U12 AP2112K -> +3V3 (every RP2354B supply pin: it
rises with +5V, so the 5 V-tolerant pads are never unpowered with 5 V on them) and U11 AP63203 -> +3V3_S3
(the ESP32-S3, the CP2102N and the microSD: WiFi bursts ~400 mA; it waits out UVLO, then soft-starts over
4 ms). Console 5 V headroom under WiFi bursts is unmeasured: bring-up step 2. TP14, TP13, TP15, TP16 and TP17 are
bring-up pads.



PWR_FLAG: these rails reach their loads only through passive parts (the edge fingers, USB-C, the buck's inductor).



Rev0 2026-10-11: RP2354B + ESP32-S3; firmware pico/studio2 s2_cart.h, fujiversal-studio2.h
CERN-OHL-W-2.0 (RP core from FujiNet-5200-Rev0, S3 half from FujiNet-ChannelF-Rev0)
Generated from tools/design.py + tools/sch_layout.py - edit those, not this file
FujiNet

Sheet: /power/
File: power.kicad_sch

Title: FujiNet Studio II Rev0 - Power: mux, S3 buck, RP LDO

Size: A3	Date: 2026-10-11	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 6/6